XIAMEN UNIVERSITY MALAYSIA

COM203 - Computer Architecture

Lab # 2

Mars



Dr. Mohammadmahdi Ariannejad

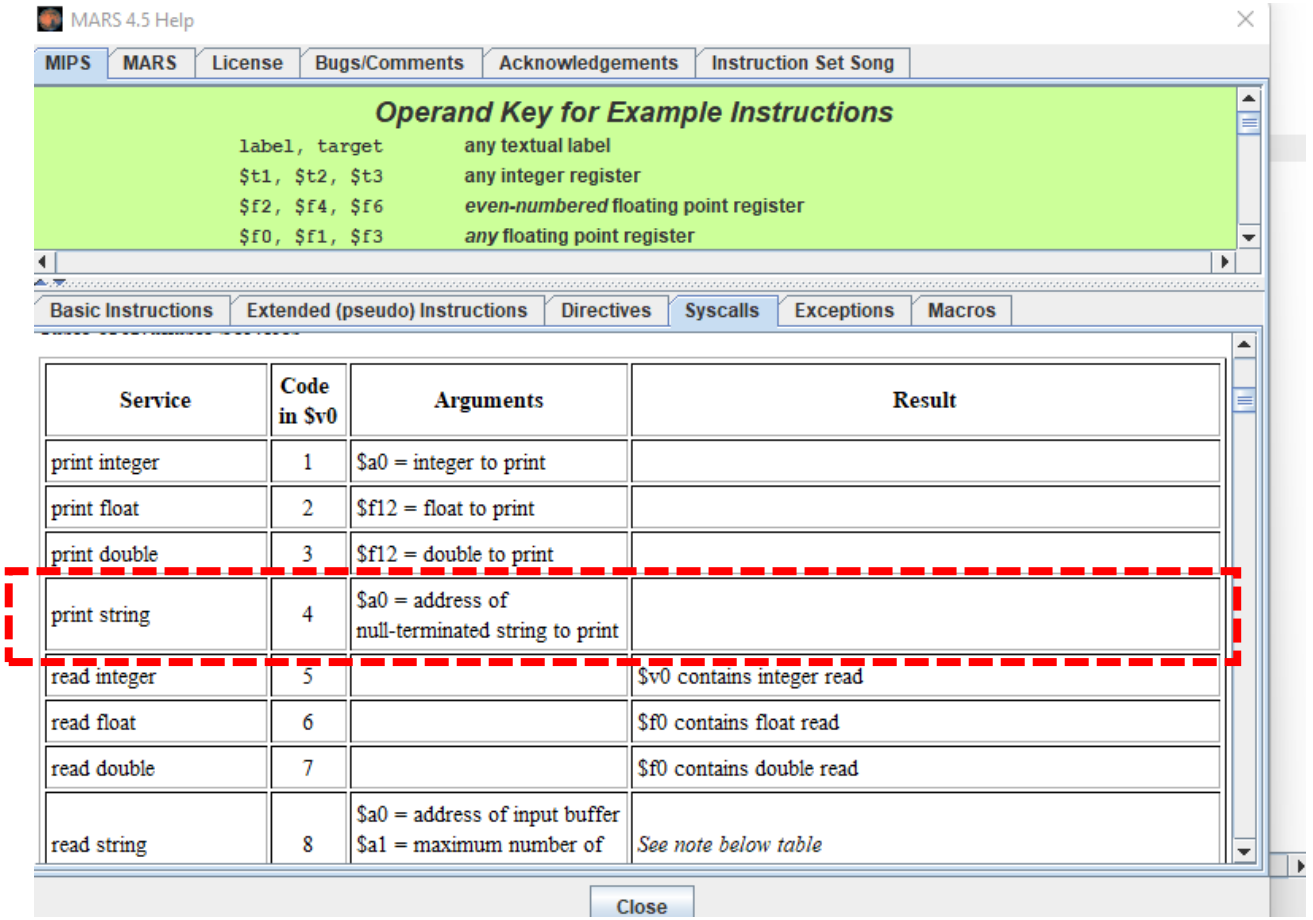
mohammadmahdi.ariannejad@xmu.edu.my

PRINTING A CHARACTER

Code:

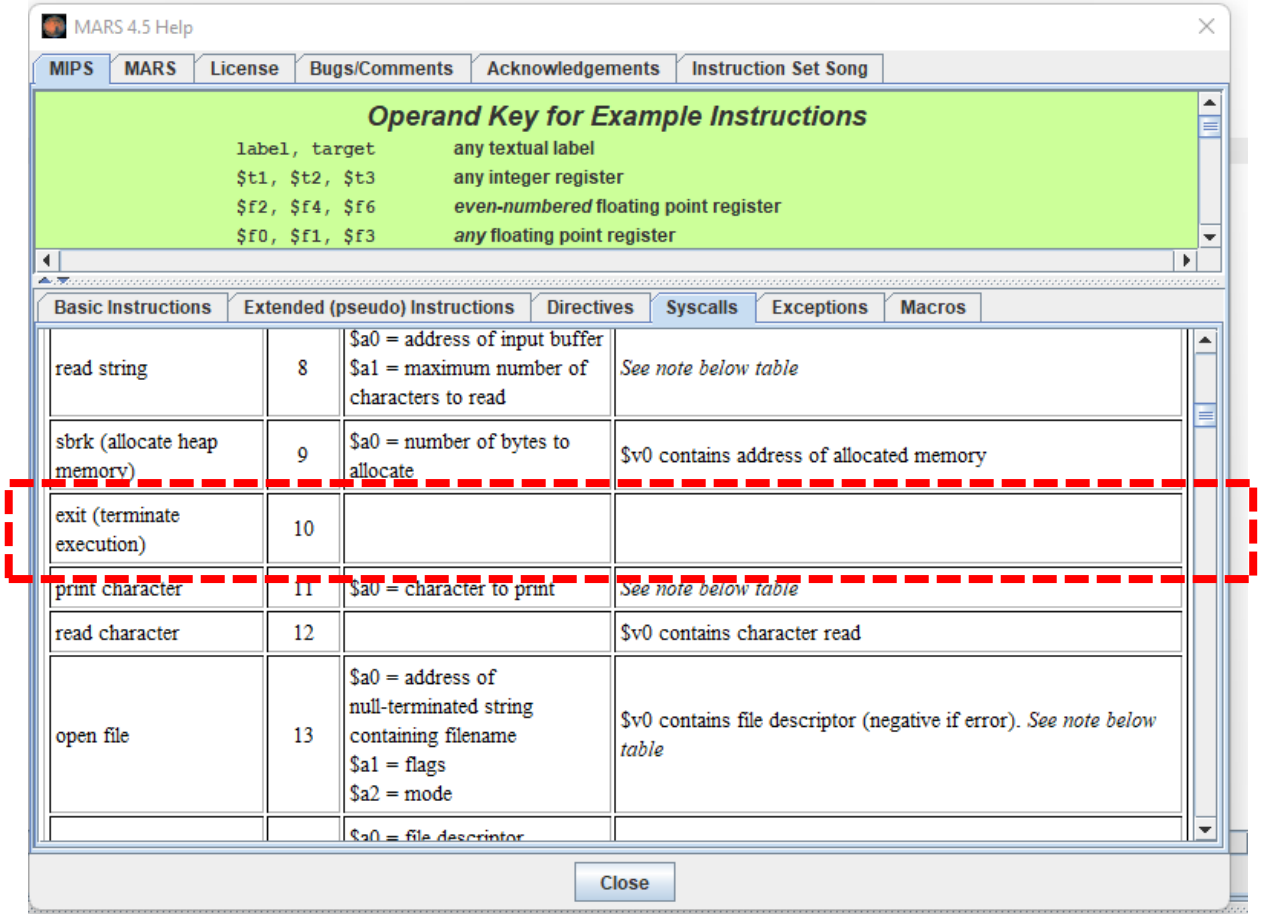
```
.data
    myCharacter: .byte 'y'
.text
    li $v0, 4                    # print command
    la $a0, myCharacter
syscall
    li $v0, 10                   #Exit command
syscall
```

Print String



Service	Code in $v0	Arguments	Result
print integer	1	$a0 = integer to print	
print float	2	$f12 = float to print	
print double	3	$f12 = double to print	
print string	4	$a0 = address of null-terminated string to print	
read integer	5		$v0 contains integer read
read float	6		$f0 contains float read
read double	7		$f0 contains double read
read string	8	$a0 = address of input buffer $a1 = maximum number of	*See note below table*

EXIT


MARS 4.5 Help
MIPS
MARS
License
Bugs/Comments
Acknowledgements
Instruction Set Song
Operand Key for Example Instructions
label, target any textual label
$t1, $t2, $t3 any integer register
$f2, $f4, $f6 even-numbered floating point register
$f0, $f1, $f3 any floating point register
Basic Instructions
Extended (pseudo) Instructions
Directives
Syscalls
Exceptions
Macros
read string 8 $a0 = address of input buffer $a1 = maximum number of characters to read See note below table
sbrk (allocate heap memory) 9 $a0 = number of bytes to allocate $v0 contains address of allocated memory
exit (terminate execution) 10
print character 11 $a0 = character to print See note below table
read character 12 $v0 contains character read
open file 13 $a0 = address of null-terminated string containing filename $a1 = flags $a2 = mode $v0 contains file descriptor (negative if error). See note below table
$a0 = file descriptor
Close